THE DEGREE OF A SPLITTING FIELD IS DIVISIBLE BY THE DEGREE OF THE SKEW FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; both steps name their input and the concluding arithmetic is displayed, so no step is left to the reader.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Splitting fields and similar algebras of prescribed degree. Same chapter, Theorem theorem-splitting. Supplies the existence, for a finite splitting field, of an algebra in the same Brauer class whose degree is the square of the degree of the splitting field.
- Similarity and matrix algebras over a skew field. Same chapter, Lemma lemma-similar. Supplies that the algebra obtained is a matrix algebra over the given skew field, which gives its degree in terms of the size of the matrix and the degree of the skew field.

Source reference. Original theorem 074Z: sources/stacks-brauer/pinned-original-source.tex, source lines 852–852. Navigation pointer invoked at proof line 903; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 074K: sources/stacks-brauer/pinned-original-source.tex, source lines 608–608. Navigation pointer invoked at proof line 917; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:brauer.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Consider a finite central skew field K over k. Let $d^2 = [K : k]$. For any finite splitting field k' for K the degree $[k' : k]$ is divisible by d.*

Proof. By Theorem 074Z there exists a finite central simple algebra B in the Brauer class of K such that $[B : k] = [k' : k]^2$. By Lemma 074K we see that $B = \mathrm{Mat}(n \times n, K)$ for some n. Then $[k' : k]^2 = n^2 d^2$ whence the result. □